

What is metaphysical necessity?

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Part A. Some unfinished business from last week.

Context: we're talking about

Iteration: Whatever is metaphysically possibly metaphysically possible is metaphysically possible

If you replace 'metaphysically' with 'ancestrally', this becomes *extremely* hard to deny. There's pressure on those who deny Iteration to escape Tolerance Puzzles to accept 'Ancestral Hypertolerance' in the corresponding ancestral-necessity puzzles. Last week I made a case against this based on certain *higher order identities* such as:

- (1) To be in the Tri-State Area is to be in New York, New Jersey, or Connecticut.
- (2) For m to be an oxygen molecule is for a_1 and a_2 to compose an oxygen molecule.
- (3) Muhammed Ali's left fist and Muhammed Ali's left hand are such that: to be hit by the former is to be hit by the latter while it is clenched.
- (4) Plato's beard is such that: to be identical to it is to be a beard of Plato.

These imply corresponding claims of ancestral necessity. And (I claimed) it's implausible that tables, pyramids, etc. are dramatically freer than areas, molecules, fists, beards as regards what's ancestrally possible for them.

1. Structure

While identities like (1–4) are *prima facie* attractive, there are possible principled reasons for rejecting them. The most influential are based on certain generalizations of:

Structure $Fx = Gy \rightarrow (x = y \wedge F = G)$

Generalized Structure

- (i) $(\lambda x_1 \dots x_k. Rx_1 \dots x_k y_1 \dots y_n \neq \lambda x_1 \dots x_k. Sx_1 \dots x_k z_1 \dots z_m \text{ (where } m=n))$
- (ii) $\lambda x_1 \dots x_k. Rx_1 \dots x_k y_1 \dots y_n = \lambda x_1 \dots x_k. Sx_1 \dots x_k z_1 \dots z_n \rightarrow R=S \wedge y_1=z_1 \wedge \dots \wedge y_n=z_n.$

2. The Russell-Myhill theorem

Super-Weak Propositional Structure: $\forall F \forall G \forall p [(F(p) = G(p)) \rightarrow \forall q (F(q) \leftrightarrow G(q))]$

This says that: for any p , "applying to p " generates a mapping from type $\langle \rangle$ to type $\langle \rangle$ in which only coextensive type- $\langle \rangle$ things are mapped to the same type- $\langle \rangle$ thing. In the light of Cantor's theorem, the following result is thus not surprising:

Russell-Myhill Theorem: *Super-Weak Propositional Structure* is inconsistent in the background logic.

Proof: Choose some arbitrary p . Let $O =_{df} \lambda q^{\diamond}. \neg \forall F((q = F(p)) \rightarrow F(q))$.

- | | |
|---|---------------------|
| (1) $O(Op) \leftrightarrow \neg \forall F(Op = Fp \rightarrow F(Op))$ | Extensional β |
| (2) $\forall F(Op \equiv Fp \rightarrow F(Op)) \rightarrow (Op = Op \rightarrow O(Op))$ | UI |
| (3) $\neg O(Op) \rightarrow (Op = Op \rightarrow O(Op))$ | (1, 2) |
| (4) $Op = Op$ | Ref |
| (5) $\neg O(Op) \rightarrow O(Op)$ | (3, 4) |
| (6) $O(Op)$ | (5) |
| (7) $\neg \forall F(Op \equiv Fp \rightarrow F(Op))$ | (1, 6) |
| (8) $\neg \forall F(Op \equiv Fp \rightarrow (O(Op) \leftrightarrow F(Op)))$ | (6, 7) |
| (9) $\neg \forall F(Op \equiv Fp \rightarrow \forall q(Oq \leftrightarrow Fq))$ | (7, UI) |

Line (9) is the denial of Super-Weak Propositional Structure. Since p was arbitrary, we can conclude that for any p , there are some non-coextensive operators F and G such that $F(p) = G(p)$.

Let me attempt at a gloss in English. Define: q is a stone-thrower iff q attributes to p some property that q lacks.

- (1): The proposition that p is a stone thrower is a stone-thrower iff it attributes to p some property that it itself lacks.
- (2): If the proposition that p is a stone-thrower doesn't attribute to p any property that it itself lacks, then if it attributes being a stone-thrower to p , it is a stone-thrower.
- (3): If the proposition that p is a stone-thrower isn't a stone-thrower, then if it attributes being a stone-thrower to p , it is a stone thrower.
- (4): The proposition that p is a stone thrower attributes being a stone-thrower to p .
- (5): If the proposition that p is a stone-thrower isn't a stone-thrower, then it is a stone thrower.
- (6): The proposition that p is a stone-thrower is a stone-thrower.
- (7): There is some property that the proposition that p is a stone-thrower attributes to p , but deoesn't have itself.
- (8): There is some property that the proposition that p is a stone-thrower attributes to p , where it's not the case that it has it itself iff it's a stone-thrower.
- (9): There is some property that the proposition that p is a stone-thrower attributes to p , which is not coextensive with (and hence not identical to) being a stone-thrower.

3. Weakenings of Structure ruled out by Beta-Identity

Atomic Structure $Fx = Gy \rightarrow (F = G \wedge x = y)$ for any *objects* x and y .

Atomic Structure (and its polyadic generalization) is consistent (in our basic background logic). However it is inconsistent with

Beta-Identity $(\lambda v_1 \dots v_n. \phi)(a_1, \dots, a_n) = \phi[a_1 / v_1, \dots, a_n / v_n]$

Example: Beta-Identity implies $(\lambda x. Lxc)(c) = Lcc = (\lambda x. Lcx)(c)$

whereas Atomic Structure implies $(\lambda x. Lxc)(c) = (\lambda x. Lcx)(c) \rightarrow (\lambda x. Lxc) = (\lambda x. Lcx)$

- This also shows the inconsistency of Atomic Structure with various weakenings of Beta-Identity, such as *Non-Vacuous Beta Identity* (whose instances are the instances of Beta-Identity where each of $v_1, \dots, v_n$ has at least one free occurrence in ϕ).

4. Weakenings of Structure consistent with Beta-Identity

Qualitative Atomic Structure $(\text{Qual}(F) \wedge \text{Qual}(G) \wedge Fx = Gy) \rightarrow (F = G \wedge x = y)$

This is consistent with Non-Vacuous Beta-Identity.

Weak Qualitative Atomic Structure

$(\text{Qual}(F) \wedge \text{Qual}(G) \wedge \forall p(F \neq (\lambda x.p) \wedge G \neq (\lambda x.p)) \wedge Fx = Gy) \rightarrow (F = G \wedge x = y)$

Finding a consistent polyadic generalization of [Weak] Qualitative Atomic Structure is tricky but possible. The following will do:

Generalized Qualitative Atomic Structure If R and S are qualitative, and objects $y_1, \dots, y_n$ are all distinct, and objects $z_1, \dots, z_m$ are all distinct, and $\lambda x_1 \dots x_k. Ry_1 \dots y_n x_1 \dots x_k = \lambda x_1 \dots x_k. Sz_1 \dots z_m x_1 \dots x_k$ then $m = n$ and for some permutation $\pi_1, \dots, \pi_n$ of the numbers $1, \dots, n$: y_1 is z_{π_1} and ... and y_n is z_{π_n} and S is $\lambda u_1 \dots u_n x_1 \dots x_k. Ru_{\pi_1} \dots u_{\pi_n} x_1 \dots x_k$.

5. Is special pleading for objects well-motivated?

Putative counterexamples to GQAS involving abstract objects:

- (1) For Obamapair to be a pair of people married to one another is for Barack Obama to be married to Michelle Obama.
- (2) For zero to number-the-children-of Wittgenstein is for Wittgenstein not to have any children.

Possible response: things like numbers and ordered pairs aren't objects. Quantification over them is "concealed higher order quantification", analogous to how we've been using words like 'property'.

Possibly reply: If this is tenable and well-motivated for abstract objects like numbers and ordered pairs, it's also tenable and well-motivated for non-fundamental objects like areas, molecules, fists, beards, tables, pyramids,

Some simple versions of this vision worth considering.

- The only real ('type- e ') objects are *particles*. Ordinary material objects are properties of particles—i.e. quantification over them is type- $\langle e \rangle$. For particle x to be part of object F is for it to be the case that Fx .

- The only real ('type- e ') objects are spacetime points. Ordinary material objects are properties of spacetime points—i.e. quantification over them is type- $\langle e \rangle$. For object F to be located at point x is for it to be the case that Fx .

Part B. Whether metaphysical necessity is absolute necessity.

An argument for Iteration:

- (1) Metaphysical necessity is the broadest necessity operation
- (2) If (1), then everything metaphysically necessary falls under every other necessity operation.
- (3) Metaphysically necessary metaphysical necessity ($\lambda p. \Box \Box p$) is a necessity operation.
- (4) So, everything metaphysically necessary is metaphysically necessarily metaphysically necessary.

Argument for premise (1): that's how the expression 'metaphysically necessary' got introduced into the philosopher's jargon. It's a descendent or equivalent of expressions like 'necessity in the highest degree — whatever that means' (Kripke 1972: p. 99); 'necessity in the widest sense' (Stalnaker 2003: p. 203); 'possibility without qualification' (van Inwagen 1998: p. 72).

- Clarke-Doane makes the following objection to this argument:

Some even come close to defining metaphysical necessity as absolute necessity (Williamson 2016). But this is unwise. The fact that metaphysical necessity is absolute is supposed to be a metaphysically significant fact. But if one simply defines it this way, then it is terminological. Given a prior understanding of "absolute", the stipulation that metaphysical necessity is absolute necessity no longer implies, for instance, that it is absolutely necessary that there are infinitely-many prime numbers, that Hesperus = Phosphorus, or that you have the parents that you have—even assuming the actual facts about numbers, Hesperus, and you.

- In response: these Kripkean claims about necessity are not supposed to follow from any *stipulation*: they are offered in a spirit of discovery and expected to be controversial. Indeed, for most of them, Kripke cares more about sweeping away what he sees as influential but bad reasons to think that they *couldn't* be true than about convincing people that they *are* true. Interpreted "metaphysical necessity" in such a way as to make their truth a matter of definition would make nonsense of this larger aim, since then there would be no interesting bad arguments against them.

Argument for premise (2): the operative notion of a “necessity operation” has to do with an operation’s *logical* behaviour. But $\lambda p.\Box\Box p$ has pretty much exactly the same logical profile as $\Box$, so if $\Box$ is a necessity operation so is $\lambda p.\Box\Box p$.

Clarke-Doane gives the following argument against (a weakening of) premise (1):

But this thesis [that metaphysical possibility is ‘the most inclusive non-epistemic, non-deontic notion of possibility’] is false. Consider logical possibility. In particular, consider the notion of logical possibility corresponding to some fixed S5 system of quantified modal logic with contingent identity Logical possibility in the present sense is dramatically more inclusive than metaphysical possibility. It is logically possible in the present sense that you could have had different parents, that Hesperus $\neq$ Phosphorus, and that there are not infinitely-many prime numbers—even assuming the actual facts about you, Hesperus, and the numbers.

- This seems to run together *predicates of sentences* with *operations*. But those are very different things!

It’s actually a lot harder to argue by counterexample that metaphysical necessity is not the broadest. Here’s one form such arguments might take (using ‘Nec’ for ‘is a necessity operation’, and O for something like ‘it is a priori that’ or what have you):

(P1) Nec(O)

(P2) $\neg O(\text{Hesperus} = \text{Phosphorus})$

(C) $\exists X(\text{Nec}(X) \wedge \neg X(\text{Hesperus} = \text{Phosphorus}))$

(C+) $\Box(\text{Hesperus} = \text{Phosphorus})$

(C++) So $\exists X \exists p(\text{Nec}(X) \wedge \Box p \wedge \neg Xp)$

The problem is that *if* both premises are granted, there is excellent reason to think that the argument is invalid. For given (P1), we can hardly deny

(P2*) $O(\text{Hesperus} = \text{Hesperus})$

But by Leibniz’s Law and the fact that Hesperus = Phosphorus, we have

(C*) $\neg \exists X(X(\text{Hesperus} = \text{Hesperus}) \wedge \neg X(\text{Hesperus} = \text{Phosphorus}))$

So the argument from (P1) and (P2) to the negation of (C*) must be invalid: this O thing creates a context in which the usual quantifier rules don’t apply. This is a common thing to think about propositional attitude verbs: consider

- (i) Hesperus = Hesperus $\wedge$ The Babylonians believed that Hesperus was visible in the evening.
- (ii) $\exists x(x = \text{Hesperus} \wedge \text{the Babylonians believed that } x \text{ was visible in the evening})$.

(iii) $\forall x(x = \text{Hesperus} \rightarrow \text{the Babylonians believed that } x \text{ was visible in the evening})$.

(iv) $\text{Phosphorus} = \text{Hesperus} \rightarrow \text{the Babylonians believed that Phosphorus was visible in the evening.}$

If you want to reject the premise and reject the conclusion there is a lot of pressure to either reject the EG step from (i) to (ii) or the UI step from (iii) to (iv).

Some say that we should accept the conclusion, or that the conclusion is true in every single context where the premise is true. But this view blocks P2 the argument for (C) above (with O as something attitudinal).

OK, so maybe we just picked a bad example? No; when we remember higher order identities, we see that the same treatment may apply to many other examples. E.g.:

(P1) $\text{Nec}O$.

(P2') $\neg O(\text{every golden thing is made of atoms containing 79 protons})$

(C') $\exists X(\text{Nec}X \wedge \neg X(\text{every golden thing is made of atoms containing 79 protons}))$

If (2) is true, we can argue as before that there is nothing we could plug in for O that makes the premises true AND makes the argument valid:

(2) To be golden is to be made of atoms containing 79 protons.